Project 3: Designing and Conducting an A/B Test

Sensitive Purchase Assistant — Human Advisor vs. Private AI

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Abstract

We compare two variants of a Shiny for Python storefront mock-up for a sensitive retail context (Mighty Patch–style acne patches): a human advisor (**A**) versus a private AI assistant (**B**). Survey outcomes (willingness, embarrassment) and behavioral metrics from Google Analytics (CTR, add-to-cart, funnel) are analyzed using the procedures summarized in this report. Treatment **B** yields large, statistically significant improvements on survey measures; behavioral differences favor **B** in magnitude but are not significant, with post-hoc power for CTR near 8%, indicating an underpowered behavioral arm rather than evidence of no effect.

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Deliverable: Final report
Live prototype: https://5243freya.shinyapps.io/stat5243-project3/
Github Repository: https://github.com/UkigawaSaeko/STAT5243-Project3
Scope: Unless stated otherwise, all statistics and procedures below describe the analysis performed for this project. Survey histograms were expanded into pseudo-individual scores for classical tests; behavioral totals come from Google Analytics exports summarized by experimental arm.

1 Introduction & Research Question

Experiments support causal inference about interventions; in applied data science, **A/B testing** is a standard framework for comparing product variants under controlled assignment. For products that may feel sensitive to discuss (for example, skincare items such as acne patches), the **tone and modality of guidance**—a visible "human advisor" versus a "private AI" assistant—could affect whether users feel comfortable engaging with help features and completing purchase-related actions.

Research question: Among visitors to a single-page **Shiny for Python** storefront mock-up, does assignment to a **private AI assistant (Treatment B)** versus a **human advisor assistant (Control A)** change (i) **self-reported willingness** to seek help, (ii) **self-reported embarrassment**, and (iii) **behavioral outcomes** (CTR, add-to-cart rate, and downstream funnel steps)?

Hypothesis: Treatment B improves comfort (**higher willingness**, **lower embarrassment**) and may improve engagement; behavioral endpoints are analyzed with explicit attention to **statistical power**.

2 Experimental Design & Methodology

2.1 Factors, arms, and assignment

- **Experimental unit:** Browser sessions on the deployed Shiny application.
- **Arms:**
  - **A (control):** Human advisor framing.
  - **B (treatment):** Private AI framing.
- **Assignment:** Uniform random assignment $P(A) = P(B) = 0.5$, implemented in the deployed web app. Assignment and session identifiers persist **only for the active browser tab**; **closing the tab and reopening** the site yields a **new independent assignment**.

2.2 Metrics

Survey (1–5 scales): willingness (higher better); embarrassment (lower better).

Behavior (Google Analytics): For each arm, impression counts define denominators for session-level rates. Assistant click and add-to-cart events define numerators for **CTR** (clicks per impression) and **add-to-cart rate** (carts per impression). Counts used here: arm A—impressions 109, clicks 52, carts 21; arm B—impressions 123, clicks 67, carts 29.

Instrumentation: Events were recorded in **Google Analytics 4** with experiment metadata and arm labels. Hosted deployments may not retain auxiliary server logs reliably; **GA4 served as the authoritative behavioral record** for aggregated rates.

2.3 Statistical methods

One pipeline ties all results together: survey frequency tables expanded to pseudo-individual scores; Welch tests and effect sizes on those scores; two-sample proportion tests on behavioral counts.

This design isolates the framing effect of assistance modality while keeping the storefront context otherwise fixed. Random assignment reduces systematic differences across users, allowing causal comparison between the two variants at the session level. We include both perceptual outcomes (willingness, embarrassment) and behavioral outcomes (CTR, add-to-cart) so that the experiment captures both psychological response and downstream engagement. A limitation is that reassignment may occur across new sessions, so inference is explicitly at the session level rather than the unique-user level.

Component	**Specification**
Means	Welch two-sample t-test (unequal variances); SciPy implementation
Robustness	Mann–Whitney U (two-sided)
Effect size	Cohen's d with pooled SD on expanded samples; $(\bar{x}_A - \bar{x}_B)$
Uncertainty	95% CI for mean differences (Welch–Satterthwaite df); willingness $\mathbf{B} - \mathbf{A}$, embarrassment $\mathbf{A} - \mathbf{B}$
Behavior	Two-proportion z-test (Statsmodels) on successes vs. impressions
Power	Post-hoc NormalIndPower for CTR using observed rate difference

3 Data Collection

Data were collected over a one-week period from April 10 to April 17, 2026. To generate sufficient traffic given the class size constraint, each team member distributed the application link through personal networks, including classmates, friends, and family. This resulted in a convenience sample of users interacting with the deployed Shiny web application under natural browsing conditions.

The survey component was embedded directly within the application and appeared at the end of the user interaction flow. Participation in the survey was entirely voluntary, and users could choose whether or not to submit responses after engaging with the assistant interface. This design ensures that survey responses reflect users who completed at least part of the interaction, though it may introduce self-selection bias, which we acknowledge as a limitation.

Survey responses were recorded as frequency tables on 1–5 Likert scales for willingness to seek help and perceived embarrassment. Invalid or incomplete responses were removed prior to analysis. To enable standard statistical procedures, frequency counts were expanded into pseudo-individual observations, which preserve the empirical distribution of responses while allowing for parametric and nonparametric testing.

Behavioral data were collected using Google Analytics 4 (GA4), which served as the authoritative source for session-level metrics. Impressions were defined as visits to the application landing page, clicks were defined as user interactions with the assistant feature, and add-to-cart events were triggered when users selected the purchase option within the interface. These definitions were consistently applied across both experimental arms. GA4 was chosen due to its reliability in tracking aggregated user behavior in hosted web environments, where server-side logging may be incomplete or inconsistent.

The combination of survey-based perceptual measures and GA4 behavioral metrics was designed to capture both psychological responses (comfort, embarrassment) and downstream en-

gagement (CTR, add-to-cart). This dual-measurement approach provides a more comprehensive evaluation of treatment effects, aligning subjective user experience with observable behavioral outcomes.

Analytic sample sizes: willingness $n_A = 49$, $n_B = 61$; embarrassment $n_A = 52$, $n_B = 58$; behavior denominators $n_A = 109$, $n_B = 123$ impressions per arm.

4 Statistical Analysis & Results

4.1 Descriptive statistics

Table 1: Willingness (1–5)

Arm	Mean	Std (sample)	Median
A	2.551	1.174	3.0
B	3.393	1.255	4.0

Table 2: Embarrassment (1–5)

Arm	Mean	Std (sample)	Median
A	3.212	1.091	3.0
B	1.931	1.024	2.0

Direction: Arm B shows higher willingness and lower embarrassment than arm A.

4.2 Inferential tests — survey

Table 3: Welch t-tests

Metric	t	p-value
Willingness	-3.627	4.44×10^{-4}
Embarrassment	6.328	6.23×10^{-9}

Both differences are statistically significant at $\alpha = 0.05$ (and well beyond conventional multiple-check thresholds).

Conclusions agree with the t-tests; results do not hinge on strict normality.

4.3 Effect sizes and confidence intervals

Cohen's d (pooled SD; $d = (\bar{x}_A - \bar{x}_B)/s_{\text{pooled}}$):
95% confidence intervals (Welch):

- **Willingness (difference B − A):** estimate **0.842**; CI **(0.382, 1.303)** — entirely above zero ⇒ B systematically higher on average.
- **Embarrassment (difference A − B):** estimate **1.281**; CI **(0.879, 1.682)** — entirely above zero ⇒ A systematically more embarrassed than B.

These intervals corroborate the significance tests.

Table 4: Mann–Whitney U (robustness)

Metric	U	p-value
Willingness	932.5	5.45×10^{-4}
Embarrassment	2401.5	3.52×10^{-8}

Table 5: Cohen's d

Metric	Interpretation
Willingness (-0.691)	Moderate-to-large effect favoring higher willingness under B
Embarrassment (1.213)	Large effect; strong practical separation on embarrassment

4.4 Behavior: rates, tests, power, funnel

Differences are **not** statistically significant at $\alpha = 0.05$; magnitudes nonetheless favor B.

Post-hoc power for CTR (Normal approximation, $\alpha = 0.05$, effect size = observed $|p_B - p_A|$):

- Effect size (rate difference) $\approx$ **0.0677**
- Estimated power $\approx$ **0.081**

This is far below the usual **0.80** target; **non-significant behavioral results are plausibly due to insufficient power** rather than proving "no effect."

Arm B is higher at CTR and slightly higher at this deeper funnel step as well, suggesting the treatment may influence more than the top of the funnel—though uncertainty remains.

4.5 Cross-metric insight — correlation

Pooling expanded scores from both arms, **Pearson correlation** between willingness and embarrassment is approximately -0.180. This negative association is consistent with a psychological mechanism: **reducing embarrassment may align with increased willingness to engage**, linking perceptual gains to the directional behavioral pattern.

5 Interpretation & Conclusion

Treatment **B** delivers **clear, statistically robust improvements** on **willingness** and **embarrassment**: large effects, significant Welch and Mann–Whitney results, and 95% CIs that exclude null shifts in the expected directions. Cohen's d supports **meaningful psychological impact**, especially for embarrassment.

Behavioral metrics (CTR, add-to-cart) **point toward B** but are **not significant** with the current impression counts. The **CTR power analysis (∼8%)** indicates the study is **not**

Table 6: Behavior rates

Metric	Rate A	Rate B
CTR	0.477	0.545
Add-to-cart / impression	0.193	0.236

Table 7: Two-proportion z-tests

Metric	z	p-value
CTR	-1.029	0.304
Add-to-cart	-0.797	0.425

Table 8: Funnel — click $\rightarrow$ add-to-cart (conditional on clicking)

Arm	Cart / click
A	0.404
B	0.433

sized to detect modest lifts; this supports interpreting the behavioral block as **inconclusive**, not as evidence that B is ineffective.

The **funnel** and **correlation** results add a coherent narrative: comfort and embarrassment move in opposing directions consistent with engagement, and early behavioral trends follow that pattern even if not yet confirmable at conventional significance.

Conclusion: Treatment B is a **strong candidate on user experience** and shows **early, directionally consistent behavioral signals**. **Follow-up experiments with larger traffic** are recommended to validate impact on CTR, cart, and downstream actions.

6 Challenges & Limitations

1. **Power:** Survey items have adequate n for detecting medium/large perceptual shifts; **behavior is underpowered** for small rate differences (quantified for CTR above).

2. **Pseudo-individual expansion:** Tests use **expanded** scores from histogram cells—appropriate for comparing arm means given published counts, but not equivalent to fully independent respondent-level rows if within-cell heterogeneity mattered.

3. **GA definitions:** Behavioral rates depend on how impressions, clicks, and add-to-cart are defined and collected; any tracking blind spot affects the rates.

4. **Session-level randomization:** Multi-device or repeat visitors are not deduplicated; inference is at the **session** level.

5. **Generalizability:** Single product narrative and convenience traffic; external validity remains open.